THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE CHAPTER TEST SET A

Chapter 6: **Control and Coordination** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The gap between two neurons is called: 1

(a) impulse **(b)** axon **(c)** dendrite **(d)** synapse

2. The part of the brain responsible for maintaining posture and balance is: 1

(a) cerebellum **(b)** medulla **(c)** cerebrum **(d)** hypothalamus

3. The plant hormone that promotes cell division is: 1

(a) ethylene **(b)** abscisic acid **(c)** cytokinin **(d)** auxin

4. A student states: “Reflex actions are controlled by the spinal cord.” The correct response is: 1

(a) Correct

(b) Only in plants

(c) Incorrect — they are controlled by the cerebrum

(d) Cannot be determined

5. For the hormone deficient in diabetes, Student A says insulin; Student B says thyroxine. Then: 1

(a) Student B is correct **(b)** Student A is correct

(c) Both are correct **(d)** Both are wrong

6. Iodine is needed by the thyroid gland to make: 1

(a) adrenaline **(b)** thyroxine **(c)** growth hormone **(d)** insulin

7. The growth of a pollen tube towards the ovule is an example of: 1

(a) hydrotropism **(b)** geotropism **(c)** chemotropism **(d)** phototropism

8. The 'fight or flight' hormone is: 1

(a) insulin **(b)** thyroxine **(c)** adrenaline **(d)** oestrogen

9. The drooping of leaves of the touch-me-not plant on touch is due to: 1

(a) gravity **(b)** growth

(c) auxin **(d)** change in water content of cells

10. Brain is protected by: 1

(a) cranium (skull) and cerebrospinal fluid **(b)** vertebral column only

(c) ribs **(d)** muscles

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 10–15 words each.

11. What is a reflex arc? 2

12. Name two plant hormones that promote growth. 2

13. Why is the use of iodised salt advisable? 2

14. A student says that hormones act at the place where they are produced. Justify or refute. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Describe the structure of a neuron with its parts and functions. 3

16. How do auxins promote bending of a shoot towards light? 3

17. Name the endocrine glands which secrete 3
(i) growth hormone
(ii) insulin
(iii) testosterone, with one function each.

OR

Differentiate between the nervous and hormonal mechanisms of coordination (three points). 3

18. What are the functions of 3
(i) medulla
(ii) cerebrum
(iii) hypothalamus?

OR

How is the timing and amount of hormone release regulated? Explain with an example. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Draw a labelled diagram (describe) of the human brain. 5
(ii) Explain the functions of the fore-brain, mid-brain and hind-brain.

OR

(i) What is a reflex action? 5
(ii) Explain the reflex arc with an example and a flow chart.
(iii) Why are reflex actions important?

20. **Case study:** A child is unusually short; another person has excessive thirst and urination with high blood sugar. 5
(i) Which gland is affected in the first case and which hormone?
(ii) Name the disorder in the second case.
(iii) Name the hormone and gland involved.
(iv) Suggest one treatment for the second case.

OR

(i) What are tropic movements? 5
(ii) Explain phototropism and geotropism with a diagram.
(iii) Give one example of hydrotropism.

END OF QUESTION PAPER — **Total: 40 Marks**

MASTERMIND